

5(a)	any three from: accept gly for glycine throughout glycine every third amino acid ; glycine, is, small / smallest, amino acid ; A has H as R-group glycine allows tight coiling of polypeptides to form triple helix ; AW A glycine presence allows a, compact / AW, triple helix gly (-NH) can form hydrogen bonds with (C=O in pro of) other polypeptides of triple helix ; A idea that H bonds can form with other polypeptides in the triple helix AVP ; e.g. pro / ala, also small amino acids (for tight coiling) high proportion of smaller amino acids / amino acids with small side chains A high proportion of glycine if mp1 <u>not</u> given and mp2 is pro can form hydroxyproline for stability ref. insoluble and (some), amino acids with non-polar R-groups (e.g. ala)	3
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Question	Answer	Marks
5(b)	any three from: (Golgi /secretory) vesicles, detail ; e.g. packaging of / containing, collagen vesicles bud off from Golgi body <i>context of movement to surface (not movement to other areas within cell)</i> - vesicles moved by, microtubules / cytoskeleton (to cell surface membrane) ; - ATP required ; A energy required / active process R active transport - fuse / merge / combine, with the <u>cell surface membrane</u> ; A <u>plasma membrane</u> - AVP ; e.g. budding from trans face of Golgi detail of fusion e.g. phospholipid bilayers merge into one	3
5(c)(i)	any two from: collagen, not broken down / not hydrolysed / breakdown prevented ; A ECM for collagen A less hydrolysis / AW detail ; e.g. suggestion that, collagen does not fit into active site / active site changes shape no / few, ESC / enzyme substrate complexes, form collagen, continues to be, synthesised / released ; A increase in collagen / maintenance of collagen	2
5(c)(ii)	(V_{max}) decreases ; (K_m) stays the same ;	2